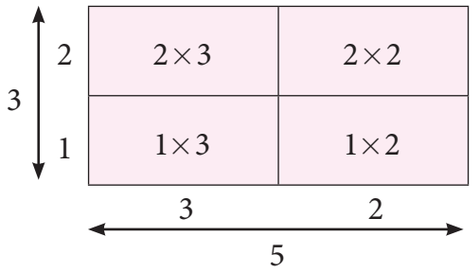
The area model helps us to understand multiplication by decomposing the area of large rectangle into areas of smaller rectangles. Also the same example which is discussed above may be decomposed as



Think

Is it the only way to decompose the numbers representing length and breadth? Discuss.

$$5\times3=(2\times3)+(2\times2)+(1\times3)+(1\times2)$$
$$=6+4+3+2$$
$$=15$$

This decomposition model is very useful when we are finding the product of large numbers.

Note

It can be practiced not to draw the area models in proportions to the numbers, since it stimulates abstract representation.

Let us try to extend this concept of multiplication for variables.

Take few tiles in rectangular shape of length 'x' and breadth 'y'. Also the area of one tile is xy (xy).

Arrange them as shown in *Fig. 3.3* and try to find its area.

xy	xy	xy
xy	xy	xy

Fig. 3.3

In *Fig. 3.3*, 6 tiles create a rectangular shape. Area of each tile is xy, hence the area of the shape is $6\times xy=6xy$. ...(1)

The same area can also be found out by taking the length of the rectangular shape (*Fig. 3.4*) as $x+x+x=3x$ and breadth as $y+y=2y$.

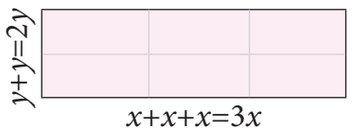


Fig. 3.4

Hence, its area = $3x\times2y$. ...(2)

Also, the same area ($6xy$) can be represented by taking the length of the rectangular shape as $y+y=2y$ and the breadh as $x+x+x=3x$.

Then the area of the rectangular shape (*Fig. 3.5*) so obtained = $2y \times 3x$. ...(3)

Since all the areas are same, from (1), (2) and (3) we get,

$$6xy = 3x \times 2y = 2y \times 3x.$$

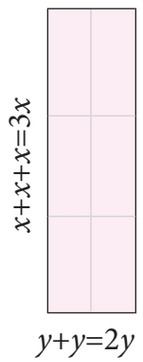


Fig. 3.5

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